STAFF REVIEW · H446 128 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. When exact reconstruction is necessary, such as executable code.

2. First in, first out.

3. An input used by a subroutine.

4. The left subtree.

I Do — source worked steps

1. AAAAA | BB | CCC — Predict the first step from the prompt.

2. (5,A)(2,B)(3,C) — Explain the intermediate result or decision.

3. Decode original — AAAAA =5A, BB=2B, CCC=3C. Encoded result (5,A)(2,B)(3,C); decoded data restores AAAAABBCCC exactly.

We Do — guided CHECK

1. A B A B A B
Predict the first step from the prompt.
Six single runs
Explain the intermediate result or decision.
Encoded size increases
Six runs of length 1 give (1,A)(1,B) repeated three times. A simple pair encoding uses 12 units for a 6-character input; RLE expands it, ignoring headers.

You Do — source mark schemes

1. (4,A)(3,B)(2,C)(4,A); one mark per correct run/length and two for correct ordered output.

2. XXXYXX [3]; order preserves original position, merging separated X runs changes sequence [2].

3. 40 runs of length one [2], count overhead expands stream [2], long blocks of identical pixels compress well [2], quantified or contextual comparison [2].

Exit Ticket — source answer

1. False. Frequent symbol changes create many short runs and can increase size.